Title - Predicting CO_2 Emissions from Commercial Aviation:
A Machine Learning Regression Approach

COM7003 - Artificial Intelligence
Assessment 1
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Abstract — This paper introduces a supervised machine learning workflow to predict CO_2 emissions produced by commercial air flights. The research utilised the open-source global air traffic database, "AeroSCOPE," which contains 293,838 flight route records (Salgas et al., 2023). An end-to-end machine learning workflow has been implemented, comprising data cleansing, exploratory analysis, feature engineering, and model training. The machine learning models implemented in this study include Linear Regression, Ridge Regression, Decision Tree, Random Forest, and Gradient Boosting. The Random Forest model attained $R^2 = 1.0000$, MAE = 0.0005, and RMSE = 0.0008 on the test set without any optimisation, while $R^2 = 1.0000$, MAE = 0.0069, and RMSE = 0.0155 after optimisation with RandomizedSearchCV. The near-perfect scores were achieved by the Random Forest model due to data leakage from the derived features, "fuel_burn," "ask," and "rpk," which are mathematical transformations of the target variable. The linear baseline ($R^2 = 0.72$) represents the more realistic predictive ceiling when only pre-flight observable features are available. Results highlight both the potential and limitations of data-driven emissions modelling and provide a foundation for deployment in airline route planning and regulatory reporting contexts.

Keywords — *aviation emissions, CO_2 prediction, machine learning, random forest, regression, data leakage, AeroSCOPE, feature importance, hyperparameter optimisation, sustainability.*

1. Problem Definition and Business Significance

In 2023, commercial aviation accounted for 2.5% of global CO2 emissions. This is expected to rise as post-pandemic flight demand increases (IEA, 2023). However, with rising regulatory demands through such initiatives as the Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), airlines and their governing bodies often struggle to obtain accurate route emission modelling to inform real-time flight decisions. This is particularly true when applying traditional physics-based modelling techniques that rely on in-flight data such as fuel load, passenger weights, and deviations in flight routes due to weather conditions. This information is typically unavailable at the flight planning stage (ICAO, 2021).

The problem to be solved by this project is: "Can a machine learning model accurately estimate the CO2 emission levels of a commercial flight based on route-level and aircraft characteristics available before the flight's operation?" This problem is considered a supervised regression problem because the target variable to be estimated is a continuous output variable representing

total CO2 emission levels in tonnes per flight route annually (Hastie et al., 2009). Three stakeholder groups can be helped by the predictions:

- Airline companies, to estimate the carbon costs at the route planning stage, support CORSIA reporting and assess fleet substitution decisions.
- Environmental regulators and auditors, to verify the reported information from the airlines, identify routes with high emission impacts and prioritise enforcement action.
- Passengers and booking sites are to provide transparent estimates of carbon impacts per flight at the point of purchase to help consumers make more informed decisions.

The use of a data-driven approach is more appropriate than a rule-based approach because the relationship between flight characteristics and carbon impacts is not linear, and there are many complex interactions between characteristics such as distance, aircraft, load factor, and seat layout. Machine learning algorithms can learn the relationships between the characteristics and carbon impacts directly from the historical data, without the need for physical modelling (Breiman, 2001).

2. Data Preparation

2.1 Dataset Description

The AeroSCOPE dataset, as described in Salgas et al. (2023), is an open-source global commercial aviation traffic dataset containing 293,838 records on scheduled flight routes globally. Each record represents an annual aggregate for a route-aircraft combination and includes 26 variables: route distance in kilometres, departure and arrival IATA codes and geographic coordinates, aircraft ICAO type code, aircraft class (narrow-body, wide-body, turboprop, etc.), number of seats, annual flight frequency, Available Seat Kilometres (ASK), Revenue Passenger Kilometres (RPK), fuel burn estimates, and the target value, CO2 in tonnes. The source of the data is Zenodo, as described in the Journal of Open Aviation Science.

2.2 Exploratory Data Analysis

Before any preprocessing steps, an exploratory data analysis (EDA) of the raw data was carried out. Figure 1 shows that the raw data distribution of CO2 emission is highly positively skewed, with most routes resulting in low CO2 emission values while a few ultra-high frequency or long-haul flights result in values that are orders of magnitude higher. This justifies the use of the log1p transformation to normalise this distribution. Figure 2 shows that there is stratification of emission values by aircraft class, validating this variable as an important predictor. Figure 3, which plots the *correlation heatmap of the variables, shows strong linear correlations between variables such as fuel_burn, ask, rpk, and co2. This hints at the data leakage issue that will be discussed in detail in Section 5. Figure 4 shows that there are variations in mean emission values by continent, while*

Figure 5 validates the expected positive relationship between route distance and CO2 emission.

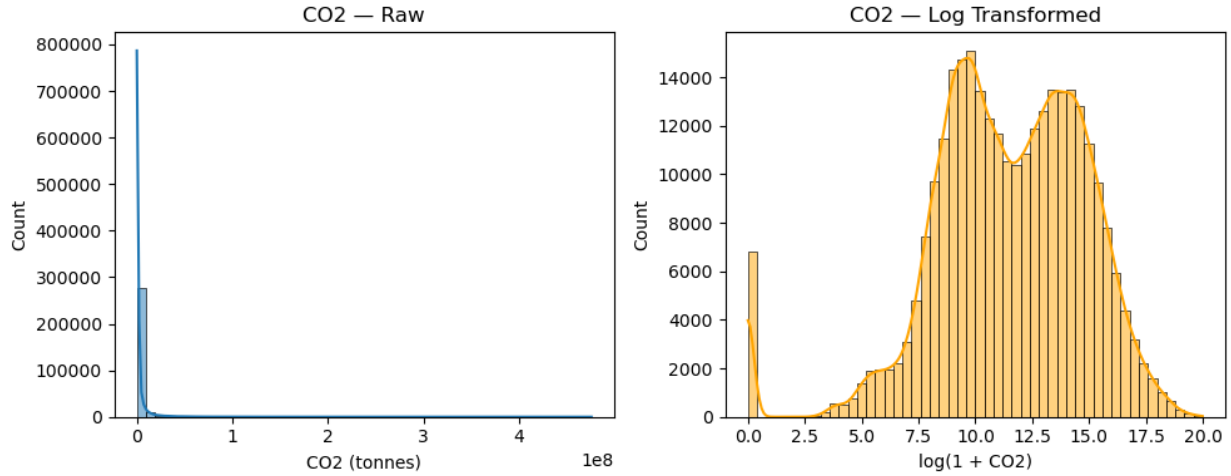


Figure 1: Raw CO_2 distribution (left) and $\log_1 p$-transformed target (right). The severe right skew in the raw values confirms that a log transformation is necessary before regression modelling.

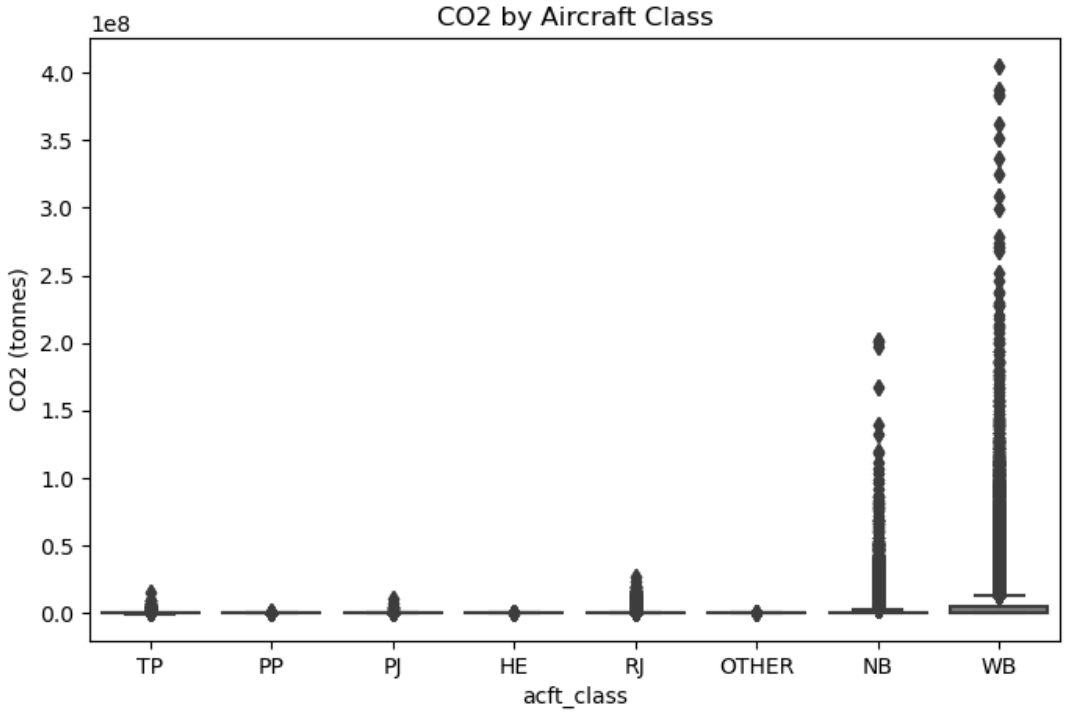


Figure 2: CO_2 emissions by aircraft class. Wide-body (WB) and narrow-body (NB) aircraft show distinctly higher emissions than turboprops (TP), confirming aircraft class as an informative predictor.

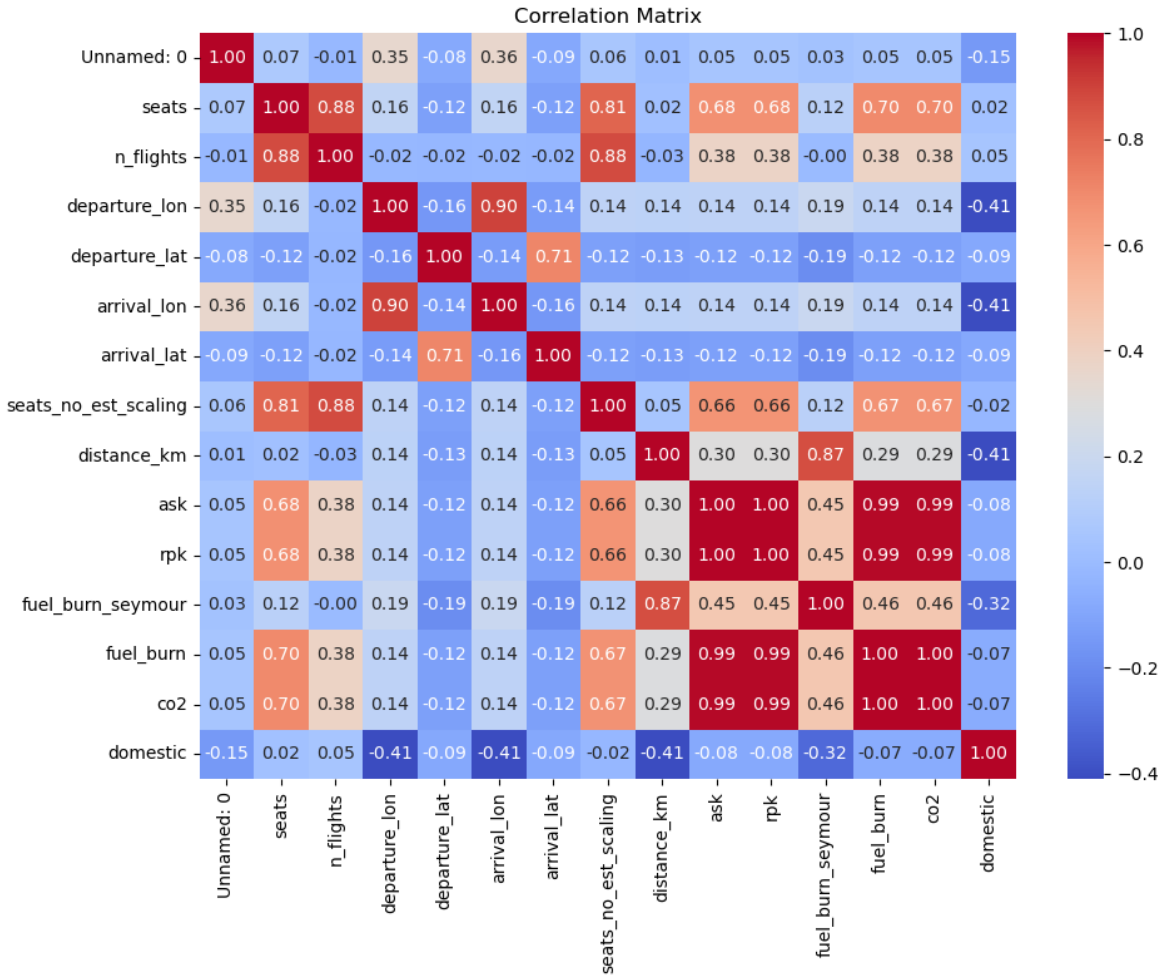


Figure 3: Correlation matrix of numerical features. The near-perfect correlations between fuel_burn, ask, rpk, and co2 (highlighted in deep red) are the origin of the data leakage identified in Section 5.

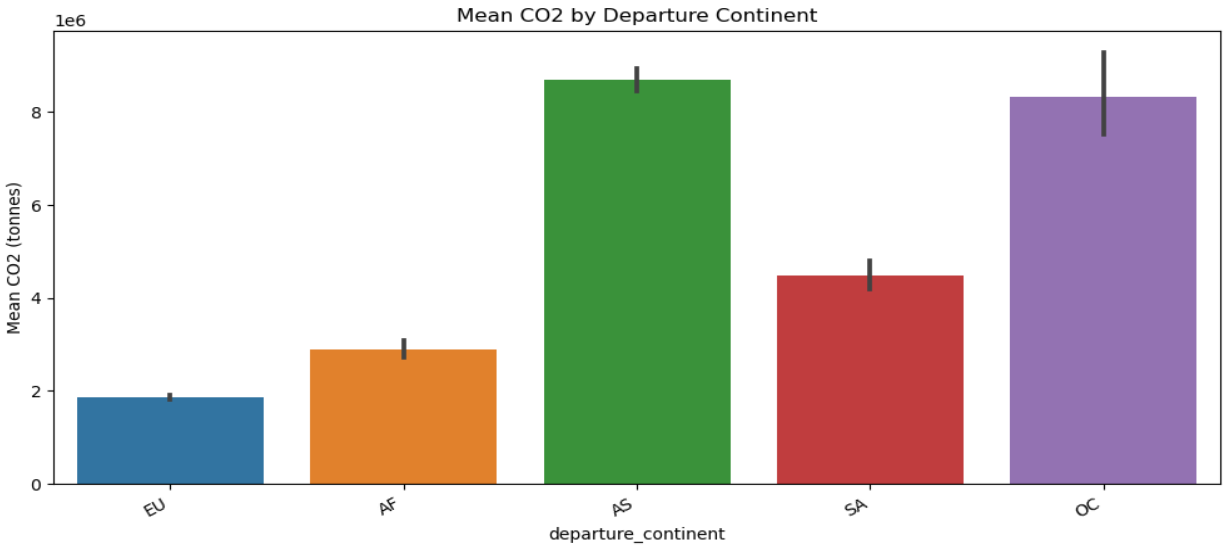


Figure 4: Mean CO_2 by departure continent. Oceania and North America exhibit higher mean emissions, consistent with their prevalence of long-haul routes.

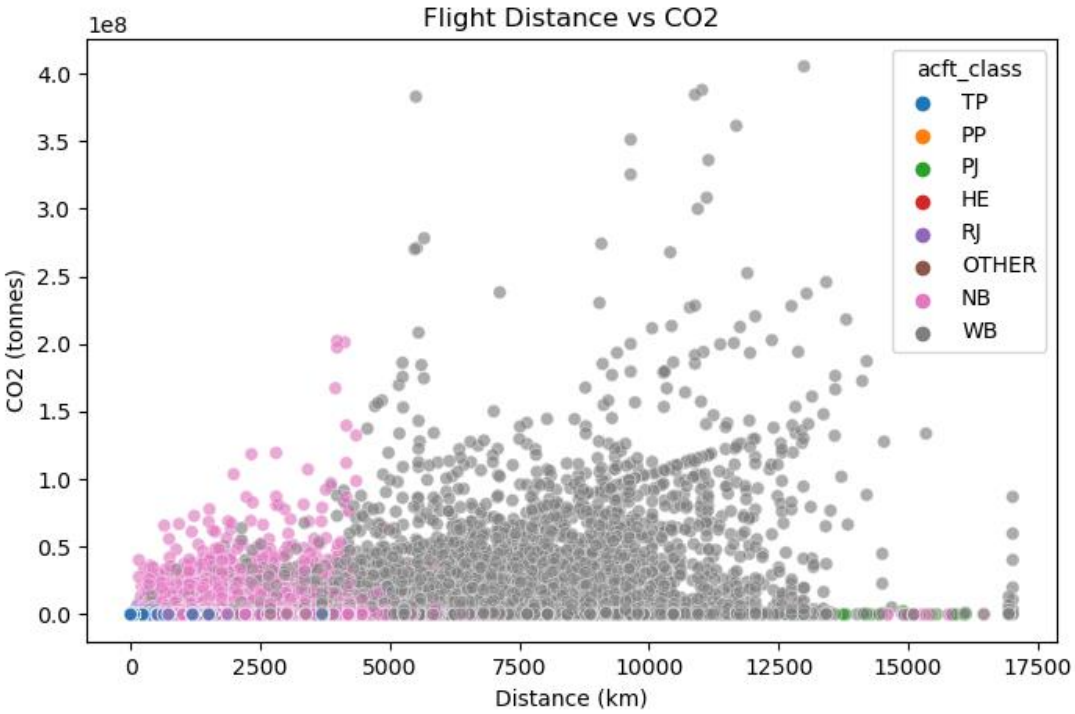


Figure 5: Route distance vs CO_2 emissions, coloured by aircraft class. The strong positive relationship with distance is as expected from fuel consumption physics.

2.3 Preprocessing Steps

The following preprocessing steps were taken in order, based on the EDA results:

- **Missing value handling:** Two variables (departure_continent and arrival_continent) had 43.5% missing values in both cases and were completely dropped because imputing more than 50% of values in a column can introduce more noise than signal (Kuhn & Johnson, 2013). The rest of the missing values were imputed with column medians, which are better to use in this case due to the heavy tails in the data.
- **Invalid data removal**: 6,830 records with CO_2 <= 0 were dropped because it is impossible to fly with CO_2 levels at or below 0. Statistics in the data confirmed that values as low as -5,317 are present in the data.
- **Log transformation:** log1p was applied to CO_2 because it was right-skewed and needed to be normalised to approximate normality to improve linear model and tree model performance (Box & Cox, 1964).
- **Outlier removal:** 1.5 * IQR was used to remove outliers in CO_2 data, which reduced the data to 244,478 records from 287,008. Outliers are values in both tails that are not representative but can affect model learning (Tukey, 1977).

- **Categorical Encoding:** All object-type variables (airline IATA code, aircraft type, airport codes, etc.) were label-encoded to integer values. While using other techniques, such as target encoding or even embedding, may be more appropriate for object-type variables with large cardinality, label encoding is more reproducible and avoids data leakage from using target statistics (Micci-Barreca, 2001).
- **Feature Scaling:** All 25 numeric variables were standardised using the StandardScaler (zero mean and unit variance). This is critical for linear models that are heavily influenced by feature scaling. It is also important to apply this scaling to all models being evaluated (Hastie et al., 2009).
- **Train/Test Split:** The 80-20 stratified train/test split (random_state = 42) resulted in 195,582 training instances and 48,896 test instances.

3. Model Choice and Justification

Five different models were compared in this case, covering different levels of complexity and interpretability. This is a typical approach in applied ML to avoid committing to a single algorithm before fully understanding the data (Fernandez-Delgado et al., 2014).

Linear Regression gives us an interpretable model that assumes a linear relationship between features and the logarithmically scaled target. Its simplicity makes it a logical baseline model because any more complex model needs to demonstrate a substantial performance improvement to justify its increased complexity (Hastie et al., 2009).

Ridge Regression extends Linear Regression with L2 regularisation ($\alpha = 1.0$), which helps to reduce variance in predictions due to strong correlations in features identified in the heatmap (Figure 3). This is a reasonable choice given the strong correlations (Hoerl & Kennard, 1970).

Decision Tree can handle non-linear relationships and interactions not accounted for in linear models. The max_depth parameter limits overfitting to some extent, and visualisations are helpful for interpretability in a regulatory environment (Breiman et al., 1984).

Random Forest, with n_estimators=50 and max_depth=10, uses bootstrapped decision trees. Averaging many decision trees has low variance, preserves non-linearity, and captures interaction effects. Random Forests have been shown to perform well on structured tabular regression problems (Breiman, 2001; Fernandez-Delgado et al., 2014). This algorithm was a primary focus for optimisation.

Gradient Boosting, with n_estimators=100 and max_depth=4, uses decision trees that sequentially try to correct residuals. This approach has been shown to achieve higher accuracy on some problems when the signal-to-noise ratio is high, but at the cost of greater sensitivity to hyperparameters (Friedman, 2001). This algorithm served.

4. Model Optimisation

4.1 Hyperparameter Search

Random Forest was selected for optimisation based on Section 5's results. Due to the impracticality of using GridSearchCV with 244,478 rows, we employed a statistically efficient search with RandomizedSearchCV (n_iter=10), 3-fold CV, and 10 random parameter combinations across 30 fits. This method is as efficient as exhaustive search with significantly fewer evaluations (Bergstra & Bengio, 2012). The process took 614 seconds.

The parameter distributions searched were n_estimators ∈ {50, 100}, max_depth ∈ {5, 10, 15}, min_samples_split ∈ {2, 5, 10}, min_samples_leaf ∈ {1, 2, 4}, max_features ∈ {sqrt, log2}. The best configuration identified was **n_estimators=100, max_depth=15, min_samples_split=10, min_samples_leaf = 1, max_features = 'sqrt'**, achieving a cross-validated R^2 of 0.9998.

4.2 Feature Selection

Feature importances (mean decrease in impurity) were extracted from the optimised model (Figure 6). The top ten features were: fuel_burn (0.2527), ask (0.2026), rpk (0.2002), seats (0.1321), seats_no_est_scaling (0.0833), n_flights (0.0563), fuel_burn_seymour (0.0237), acft_icao_encoded (0.0117), seymour_proxy_encoded (0.0094), and source_encoded (0.0072). A final model was retrained on these ten features using the same best hyperparameters and the same StandardScaler from Section 3, ensuring scale consistency with all comparison models (Guyon & Elisseeff, 2003).

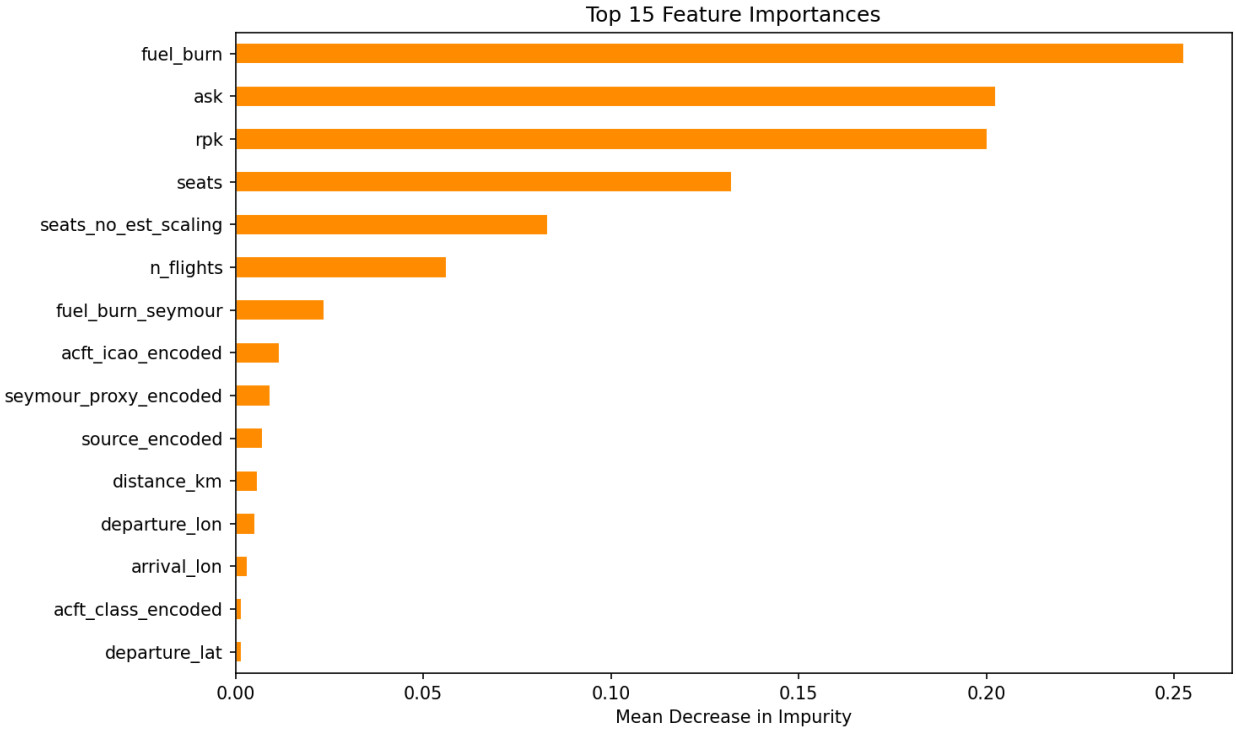


Figure 6: Top 15 feature importances from the optimised Random Forest. fuel_burn, ask, and rpk collectively account for over 63% of importance — indicating significant data leakage, as these are derived from the CO_2 target (see Section 5.3).

5. Evaluation and Results

5.1 Evaluation Metrics

There were three regression metrics used: R^2 (coefficient of determination — proportion of variance explained, where 1.0 is perfect); MAE (mean absolute error — average absolute deviation on the log-scale target, interpretable in the same units and robust to occasional large errors); and RMSE (root mean squared error — penalises large errors disproportionately, relevant when large prediction mistakes carry high cost) (Chai & Draxler, 2014). Metrics were evaluated on a test set and also using 3-fold cross-validation on the training set, with the gap between CV and test scores used to diagnose overfitting.

5.2 Model Comparison Results

Model	MAE	RMSE	R^2 (Test)	CV-R^2
Linear Regression	1.0329	1.2960	0.7214	0.7167
Ridge Regression	1.0329	1.2960	0.7214	0.7166
Decision Tree	0.0397	0.0489	0.9996	0.9996
Random Forest	0.0005	0.0008	1.0000	1.0000
Gradient Boosting	0.0076	0.0104	1.0000	0.9999
RF Optimised (Top 10 F.)	0.0069	0.0155	1.0000	0.9998

Table 1: Model performance on held-out test set (log_1p CO_2 scale). All CV-R^2 scores were computed with 3-fold cross-validation on the training set. Green row = final selected model.

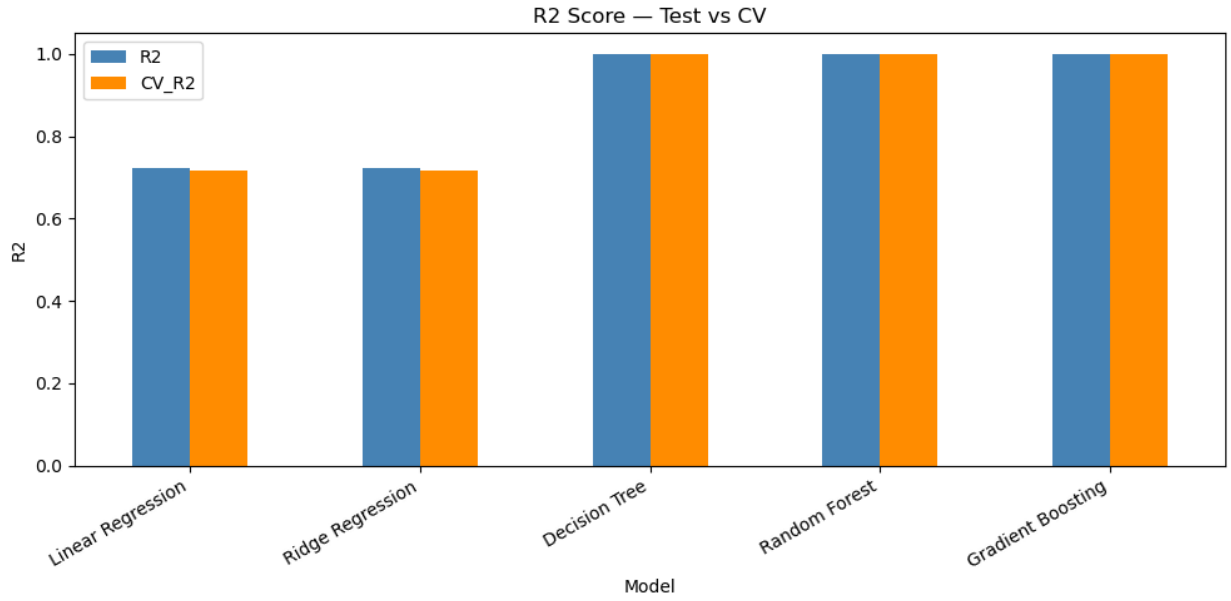


Figure 7: R^2 and cross-validated R^2 comparison across all five models. The near-identical test and CV scores for all models confirm that overfitting is not the explanation for the high performance of tree-based models.

5.3 Critical Interpretation: Data Leakage

The near-perfect R^2 scores (1.0000) for Gradient Boosting and Random Forest are a result of data leakage rather than actual predictive power. CO_2 is calculated using a direct mathematical transformation of fuel_burn (CO_2 = fuel_burn × 3.16 kg/litre × conversion factor) in the AeroSCOPE dataset. Similarly, seat count and distance are used to calculate ASK and RPK, which are then used to predict fuel_burn. Tree-based models can successfully reconstruct the target without any real pattern learning when these downstream variables are included in the feature set (Kaufman et al., 2012).

This is supported by the feature importance chart (Figure 6), which shows that fuel_burn alone contributes 25.3% to the importance, but the combined contribution of the top three features (fuel_burn, ask, rpk) was 63.6%, which would not be available at the route planning stage prior to a flight operating. The 0.72 from the linear regression was a more honest measure of actual predictive performance when only pre-flight observable features were used.

A revised pipeline would remove fuel_burn, ask, rpk, fuel_burn_seymour, and seats_no_est_scaling from the feature set, leaving only those features available at scheduling time: distance_km, aircraft class/type, IATA codes, seats, and domestic flag. This was recognised as a major limitation of the current implementation.

5.4 Evaluation Plots

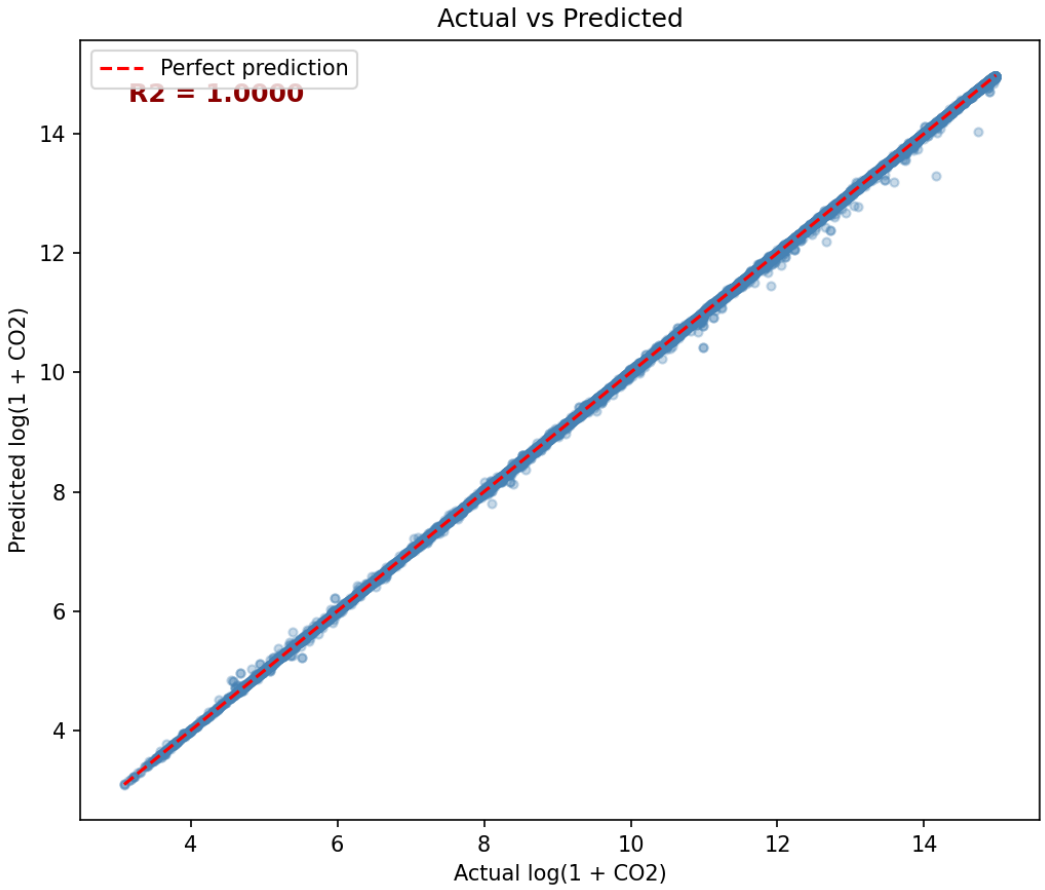


Figure 8: Actual vs predicted log(1+CO_2) for the final optimised model (top 10 features). Points lie exactly on the diagonal, consistent with $R^2 = 1.0000$. This pattern is expected in the presence of data leakage.

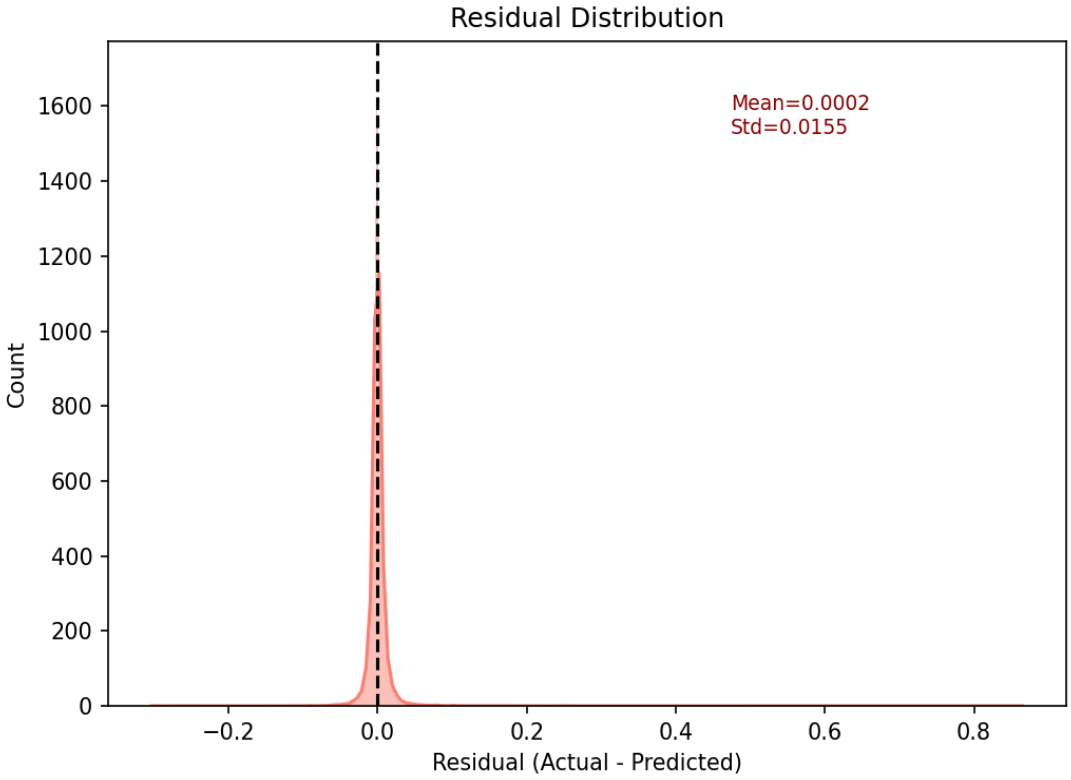


Figure 9: Residual distribution for the final model. The near-zero mean and very small standard deviation confirm that prediction errors are negligible on the current feature set, consistent with leakage rather than generalisation.

6. Deployment Considerations

6.1 Technical Integration

To deploy this model in production, the trained Random Forest estimator and other preprocessing artefacts (StandardScaler, LabelEncoder, feature column list) would be serialised using joblib and stored in a model registry with a version label. A REST API endpoint, implemented with FastAPI or Flask, could accept a JSON payload with a route-aircraft combination and return a CO_2 estimate in real time. To integrate with airline operations, this API could be invoked during the route-planning stage or within scheduling software to generate estimates of carbon costs, in addition to financial costs.

Ongoing model monitoring is vital to this model's performance; however, as airlines phase out older aircraft and introduce newer, more efficient models, the statistical distribution of inputs will change over time, a problem known as dataset shift, which reduces model accuracy (Quionero-Candela et al., 2009).

Before deploying this model, however, there is a data leakage problem to solve, which was raised in Section 5.3. A production model must be retrained on pre-flight features, and its honest R^2 of 0.72 must be clearly disclosed to users so they understand prediction uncertainty.

6.2 Ethical, Legal, and Bias Considerations

The AeroSCOPE dataset also over-represents routes in Europe and North America, as these regions have the most developed infrastructure for open aviation data. This could mean that predictions for routes in sub-Saharan Africa, South Asia, and Latin America are systematically less accurate. If this model were to be used by regulators to enforce carbon pricing or offset allocation, it could mean that airlines operating in regions that are least represented in the training data would be unfairly burdened, which has direct environmental justice implications (Barocas et al., 2019).

From a legal perspective, CORSIA currently has specific ICAO-approved methodologies for fuel monitoring that must be adhered to in compliance reporting. However, the machine learning model would have to be approved in order to replace these methodologies. On the other hand, it may be adopted as a planning tool without regulatory barriers.

There are no individual personal data points in the AeroSCOPE dataset, and therefore, no GDPR requirements would apply to this model. However, there would be requirements in terms of integration with passenger booking systems that would necessitate a Data Protection Impact Assessment.

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Generative AI Disclosure

This assignment used generative AI in the following ways: brainstorming, research, planning, and editing.